MTH101: Calculus and Analytical Geometry

Midterm Formula Sheet (Lessons 1 - 22)

1. Coordinates and Lines (Lessons 1-4)

- Slope (m): $m = (y_2 - y_1) / (x_2 - x_1)$
- Slope-Intercept Form: $y = mx + b$
- Point-Slope Form: $y - y_1 = m(x - x_1)$
- General Equation: $Ax + By + C = 0$
- Parallel Lines: $m_1 = m_2$
- Perpendicular Lines: $m_1 \times m_2 = -1$

2. Distance and Circles (Lesson 5)

- Distance Formula: $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- Midpoint Formula: $M = ((x_1+x_2)/2 , (y_1+y_2)/2)$
- Standard Circle Equation: $(x - h)^2 + (y - k)^2 = r^2$

3. Limits and Continuity (Lessons 9-13)

- Special Trig Limit: $\lim_{x \to 0} (\sin x / x) = 1$
- Continuity Condition: $\lim_{x \to c} f(x) = f(c)$

4. Differentiation Rules (Lessons 15-20)

Rule Name	Formula
Power Rule	$d/dx(x^n) = nx^{n-1}$

Product Rule	$(fg)' = f'g + fg'$
Quotient Rule	$(f/g)' = (f'g - fg') / g^2$
Chain Rule	$dy/dx = (dy/du) \times (du/dx)$

Trigonometric Derivatives

- $d/dx(\sin x) = \cos x$
- $d/dx(\cos x) = -\sin x$
- $d/dx(\tan x) = \sec^2 x$
- $d/dx(\sec x) = \sec x \tan x$

Exponentials and Logarithms

- $d/dx(e^x) = e^x$
- $d/dx(\ln x) = 1/x$

5. Applications (Lessons 21-22)

- **Newton's Method:** $x_{n+1} = x_n - f(x_n) / f'(x_n)$
- **Tangent Line Equation:** $y - f(x_0) = f'(x_0)(x - x_0)$